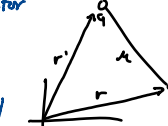


- 1) r is the field point position vector
origin $\rightarrow$ eval point: 
- 2) r' is the source point position vector
origin $\rightarrow$ source: 
- 3) r is the separation vector. $r = |r - r'|$


Electrostatics

Field and Force:

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

Discrete: $E(r) = \frac{1}{4\pi\epsilon_0} \sum \frac{q_i}{r_i^2} \hat{r}_i$

Continuous: $E(r) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(r')}{r^2} \hat{r} d\tau'$

Flux and Gauss's Law:

$$\oint E \cdot da = Q_{enc}/\epsilon_0$$

over a closed surface (Gaussian Surface)

Other way to find E, but

$Q_{enc} = \int \rho d\tau$

$\nabla \cdot E = \rho/\epsilon_0$

$\nabla \cdot E = 0$

Potential and Energy:

$$\Delta V = -\nabla^2 V = -\rho/\epsilon_0$$

In a closed vol, $\Delta V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r} = 0$

Stokes': $\oint A \cdot dl = \int \nabla \times A \cdot \hat{n} dA$

$\Rightarrow \oint E \cdot dl = \int \nabla \times E \cdot \hat{n} dA = 0$

$\nabla \times E = 0$

$E = -\nabla V$ and $\nabla \cdot E = \rho/\epsilon_0$

$\Rightarrow \nabla^2 V = -\rho/\epsilon_0$

$\nabla^2 V = 0$

$V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(r')}{r} d\tau'$

$W = Q[V(b) - V(a)]$

Discrete: $W = \frac{1}{2} \sum q_i V(r_i)$

In sphere: $W = \frac{1}{2} \int \rho V d\tau$, $\rho \neq 0$

For any V enclosing the charge: $W = \frac{\epsilon_0}{2} \left(\int E^2 d\tau + \oint V E \cdot da \right)$

All space: $W = \frac{\epsilon_0}{2} \int E^2 d\tau$

Conductors and Capacitors:

$E = 0$ inside

$\rho = 0$ inside

Any net charge resides on the surface

$V(a) = V(b)$ $\forall a, b$ inside

E is perpendicular to the surface just outside, i.e. $E = \frac{\sigma}{\epsilon_0} \hat{n}$

$\sigma = -\epsilon_0 \partial V / \partial n$

f (force per unit area) $= \frac{\sigma^2}{2\epsilon_0} \hat{n}$

Outward electrostatic pressure: $P = \frac{\epsilon_0}{2} E^2$

Capacitance: $C \equiv Q/V$ (F)

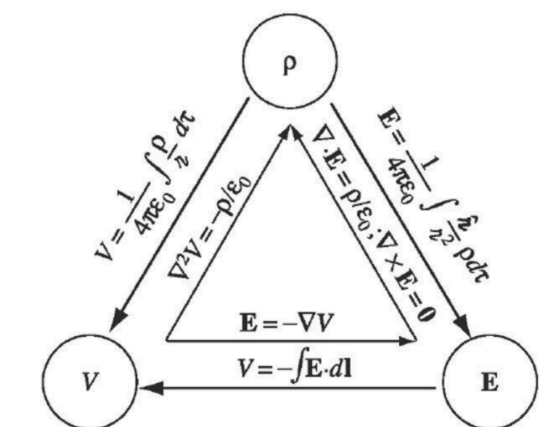
$W = CV^2/2$

Boundary Conditions:

Below $-E_{below} = \sigma/\epsilon_0$

Above $= V_{below}$

$\partial V / \partial n - \partial V_{below} / \partial n = -\sigma/\epsilon_0$



Potentials

Cartesian Separation of Variables:

$$\nabla^2 V = 0$$

$$\partial^2 V / \partial x^2 + \partial^2 V / \partial y^2 + \partial^2 V / \partial z^2 = 0$$

Force on a dipole: $F = (p \cdot \nabla) E$

V polarized: $V(r) = \frac{1}{4\pi\epsilon_0} \int \rho(r') \frac{1}{r} d\tau'$ with $\rho_b = -\nabla \cdot P$ and $\sigma_b = P \cdot \hat{n}$

Electric Displacement: $D = \epsilon_0 E + P$

Free charge: $\nabla \cdot D = \rho_f$

$\oint D \cdot da = Q_{fenc}$

$\nabla \times D = 0$

Dipole Field in Spherical Coords: $E(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2\cos\theta \hat{r} + \sin\theta \hat{\theta})$

Dipole Field Coord-free: $E(r) = \frac{1}{4\pi\epsilon_0 r^3} [3(p \cdot \hat{r}) \hat{r} - p]$

Spherical Separation of Variables:

$\rho = 0$: $\partial^2 V / \partial r^2 + \frac{1}{r^2} \partial^2 V / \partial \theta^2 + \frac{1}{r^2 \sin^2 \theta} \partial^2 V / \partial \phi^2 = 0$

$V(r, \theta) = R(r) \Theta(\theta)$

$\frac{1}{R} \frac{d}{dr} (r^2 \frac{dR}{dr}) + \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} (\sin \theta \frac{d\Theta}{d\theta}) = 0$

$\frac{l(l+1)}{R} = -\frac{1}{\Theta} \frac{d}{d\theta} (\sin \theta \frac{d\Theta}{d\theta})$

$R(r) = A r^l + B / r^{l+1}$, $\Theta(\theta) = P_l(\cos \theta)$

$V(r, \theta) = \sum (A_l r^l + B_l / r^{l+1}) P_l(\cos \theta)$

Trick: $\int_0^\pi P_l(\cos \theta) P_{l'}(\cos \theta) \sin \theta d\theta = \frac{2}{2l+1} \delta_{ll'}$ if $l = l'$

$A_l = \frac{2l+1}{2\pi} \int_0^\pi V(\theta) P_l(\cos \theta) \sin \theta d\theta$

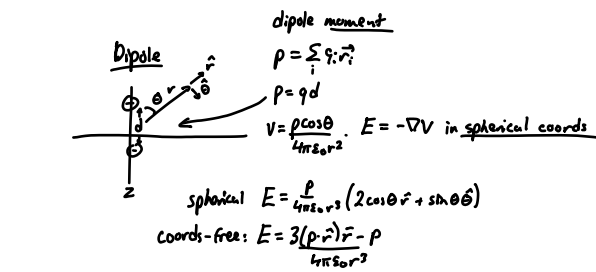
$B_l = \frac{2l+1}{2\pi} \int_0^\pi V(\theta) P_l(\cos \theta) \sin \theta d\theta$

Multipole Expansion:

$$V(r) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos \alpha) \rho(r') d\tau'$$

$V(r) = \frac{1}{4\pi\epsilon_0} \left[\frac{1}{r} \int \rho(r') d\tau' + \frac{1}{r^2} \int r' \cos \alpha \rho(r') d\tau' + \dots \right]$

monopole dipole



Electric Fields In Matter

Dielectrics and Polarization:

$P = qd = \alpha E$

Torque: $N = qd \times E = p \times E$

Force on a dipole: $F = (p \cdot \nabla) E$

V polarized: $V(r) = \frac{1}{4\pi\epsilon_0} \int \rho(r') \frac{1}{r} d\tau'$ with $\rho_b = -\nabla \cdot P$ and $\sigma_b = P \cdot \hat{n}$

Electric Displacement: $D = \epsilon_0 E + P$

Free charge: $\nabla \cdot D = \rho_f$

$\oint D \cdot da = Q_{fenc}$

$\nabla \times D = 0$

Dipole Field in Spherical Coords: $E(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2\cos\theta \hat{r} + \sin\theta \hat{\theta})$

Dipole Field Coord-free: $E(r) = \frac{1}{4\pi\epsilon_0 r^3} [3(p \cdot \hat{r}) \hat{r} - p]$

Boundary Conditions:

$D_{ab}^+ - D_{ab}^- = \sigma_f$

$E_{ab}^+ - E_{ab}^- = \sigma_b / \epsilon_0$

$E_{ab}^+ - E_{ab}^- = 0$

Linear Dielectrics:

$P = \epsilon_0 \chi_e E$ (χ_e is proportional to field)

$D = \epsilon_0 (1 + \chi_e) E$

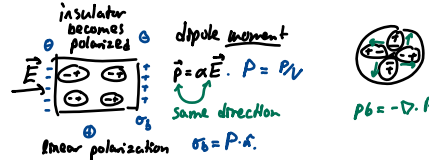
$\epsilon = \epsilon_0 (1 + \chi_e)$ when $D = \epsilon E$

$\epsilon_r \equiv 1 + \chi_e = \epsilon / \epsilon_0$ is relative permittivity

$W = CV^2/2$

$C = \epsilon_r C_{vac}$ for a capacitor filled with dielectric material

In dielectrics: $W = \frac{\epsilon_0 \epsilon_r}{2} \int E \cdot E d\tau = \frac{1}{2} \int D \cdot E d\tau$



$\nabla \cdot E = \rho / \epsilon_0 = \rho_b / \epsilon_0$

$\nabla \cdot \epsilon E = \rho_b + \rho_f$

$\nabla \cdot \epsilon_r E = -\nabla \cdot P + \rho_f$

$\nabla \cdot (\epsilon_r E + P) = \rho_f$

$D = \epsilon_0 E + P \Rightarrow \nabla \cdot D = \rho_f$

$\oint D \cdot da = Q_{fenc}$

Magnetostatics

The Lorentz Force Law:

$F_{mag} = Q(v \times B)$

Lorentz Force Law: $F = Q[E + (v \times B)]$

Cyclotron Motion: $p = qBR$

Cyclotron Motion: $(y - R \sin \omega t)^2 + (z - R)^2 = R^2$

Cyclotron Motion: $y = R \sin \omega t$, $z = R$, $u = E/B$

F_{mag} on current: $F_{mag} = \int I (dl \times B)$

F_{mag} w/ $J = \rho v$: $F_{mag} = \int (J \times B) d\tau$

F_{mag} w/ $k = \sigma v$: $F_{mag} = \int (K \times B) da$

$\nabla \cdot J = -\partial \rho / \partial t$

The Biot-Savart Law

$B(r) = \frac{\mu_0}{4\pi} \int \frac{J(r') \times \hat{r}}{r^2} d\tau'$ (T)

Circular Loop: $B(z) = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}$

$\nabla \cdot B = 0$, $\nabla \times B = \mu_0 J$

$\oint B \cdot dl = \mu_0 I_{enc}$

Magnetic Vector Potential:

$B = \nabla \times A$, with A = vector potential

$\nabla \cdot A = 0$

$A(r) = \frac{\mu_0}{4\pi} \int \frac{J(r')}{r} d\tau'$

Boundary Conditions:

$B_{ab}^+ - B_{ab}^- = 0$

$B_{ab}^+ - B_{ab}^- = \mu_0 k$

$B_{ab} - B_{ba} = \mu_0 (k \times \hat{n})$

$A_{ab} = A_{ba}$

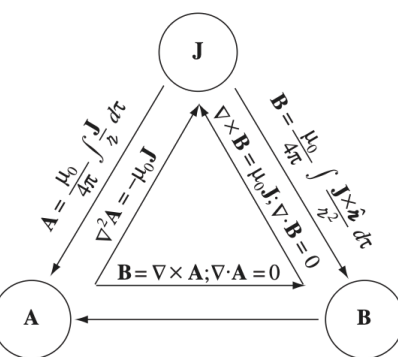
$\partial A_{ab} / \partial n - \partial A_{ba} / \partial n = -\mu_0 k$

Multipole Expansion:

$A(r) = \frac{\mu_0}{4\pi} \sum_{n=1}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos \alpha) J(r') d\tau'$

Dipole: $A_{dip}(r) = \frac{\mu_0}{4\pi} \frac{m \times \hat{r}}{r^2}$, with $m \equiv \int r' da = I a$

$B_{dip}(r) = \frac{\mu_0}{4\pi r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta})$ or $= \frac{\mu_0}{4\pi r^3} [3(\hat{m} \cdot \hat{r}) \hat{r} - \hat{m}]$



Magnetic Forces In Matter:

Torques And Forces:

Torque on Dipole: $N = m \times B$

Force on Dipole: $F = \nabla(m \cdot B)$

Magnetization and Bound Currents:

Magnetization: $M = m/V$

= magnetic dipole moment / unit volume

Bound Volume Current: $J_b = \nabla \times M$

Bound Surface Current: $K_b = M \times \hat{n}$

$A = \frac{\mu_0}{4\pi} \int J_b / r d\tau + \frac{\mu_0}{4\pi} \int K_b / r da$

Auxiliary Field H:

Total Current: $J = J_b + J_f$

$H = \frac{B}{\mu_0} - M$

$\nabla \times H = J_f$

$\oint H \cdot dl = I_{fenc}$

$\nabla \times B / \mu_0 = \nabla \times M + J_f$

$J_f = \nabla \times (B / \mu_0 - M)$

$J_f = \nabla \times H \Rightarrow B = \mu_0 (H + M)$

Boundary Conditions:

$B_{ab}^+ - B_{ab}^- = 0$

$B_{ab}^+ - B_{ab}^- = \mu_0 (k \times \hat{n})$

$H_{ab}^+ - H_{ab}^- = -(M_{ab}^+ - M_{ab}^-)$

$H_{ab}^+ - H_{ab}^- = k_f \times \hat{n}$

$\nabla \cdot H = -\nabla \cdot M$

Linear And Nonlinear Media:

$M = \chi_m H$. In paramagnets, $\chi_m > 0$. In diamagnets, $\chi_m < 0$

Magnetic Permeability: $\mu = \mu_0 (1 + \chi_m)$

Electrodynamics:

Ohm's Law:

Current density proportional to force per unit charge: $J = \sigma E$

σ is conductivity. $\rho = 1/\sigma$ is resistivity.

$J = \sigma(E + v \times B)$. Since velocities are low, we can neglect v : $J = \sigma E$ (Ohm's Law)

Electromotive Force:

or emf: $\mathcal{E} = \oint f \cdot dl$ with $f = f_b + E$ (source + electric force)

$V = - \int_a^b E \cdot dl = \int_a^b f_b \cdot dl = \oint f_b \cdot dl = \mathcal{E}$ (work/unit charge by source)

Generator: $\mathcal{E} = \oint f_{mag} \cdot dl = v B h$

$\Phi = \int B \cdot da = B h x$ As $x \rightarrow \frac{d\Phi}{dt} = B h \frac{dx}{dt} = -B h v = \mathcal{E}$

Faraday's Law: $\nabla \times E = -\partial B / \partial t$

Mutual Inductance: $\Phi = \int B \cdot da = M_{21} I_1$, with $M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{dl_1 \cdot dl_2}{r}$

Self Inductance: A changing current induces an emf in the source loop itself: $\Phi = L I$, with L = self inductance and emf $\mathcal{E} = -L dI/dt$

Magnetic Potential Energy stored in an inductor: $W = L I^2 / 2$

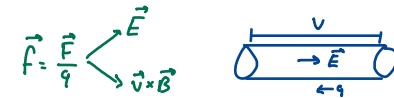
Maxwell's Equations:

$\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \partial E / \partial t$ (A changing electric field induces a magnetic field)

with a new term $J_d = \epsilon_0 \partial E / \partial t$ called the displacement current

In a vacuum, $\nabla \cdot E = 0$ and $\nabla \cdot B = 0$. By applying curl to $\nabla \times E = -\partial B / \partial t$ and $\nabla \times B = \mu_0 \epsilon_0 \partial E / \partial t$:

$\nabla^2 E = \mu_0 \epsilon_0 \partial^2 E / \partial t^2$ and $\nabla^2 B = \mu_0 \epsilon_0 \partial^2 B / \partial t^2$



$J = I/A = \sigma E = \sigma V/L$

$I = \frac{\sigma A}{L} V = \frac{V}{R} \Rightarrow V = IR$ (Ohm's Law)

with $\sigma A/L = 1/R$ or $\rho A/L = R$